Sample Midterm Exam 2

Part 1 – New Stuff

1. a) What is the main advantage of binary search over sequential search?

b) What is the main advantage of sequential search over binary search?

2. Write one line of code that dynamically allocates space for an array of doubles of size 250, and sets its reference to a new variable called **arr**.

3. Draw the "box and arrow" pointer diagram resulting from executing the below code.

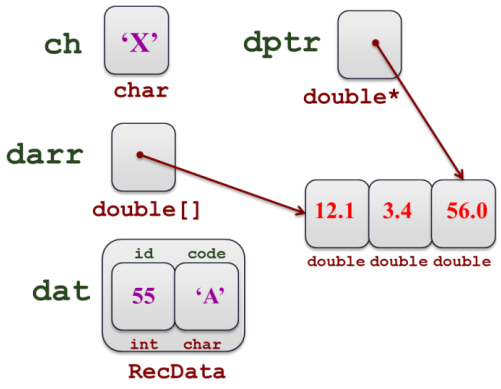
```
int *a, *b, c = 5, d = 6;
double e, f[3] = { 1.0, 2.0, 3.0 }, *g;
int **z;

a = &c;
b = a;
z = &b;
*b = 1;
a = &d;
b = &c;

e = 10.0;
*(f + 1) = e;
e = 0.0;
g = &f[2];
*g = 3.14;
```

4. Write some code that will result in the pointer diagram shown below. Do not forget that you first need to declare the variables `ch`, `darr`, `dat`, and `dptr`.

```
typedef struct {
     int  id;
     char code;
} RecData;
```



5. For parts (a), (b) and (c) below, you are not allowed to use functions from the <stdlib.h> library.

a) Implement the function `charCount`, which counts the number of occurrences of the char `ch` in the string `myStr`.

```
int charCount(char myStr[], char ch) {
```

b) Implement the function `strCat`, which concatenates `endStr` to the end of `frontStr`. Here, `frontStr` is changed, while `endStr` remains the same. We assume that `frontStr` is large enough to store the resulting string value.

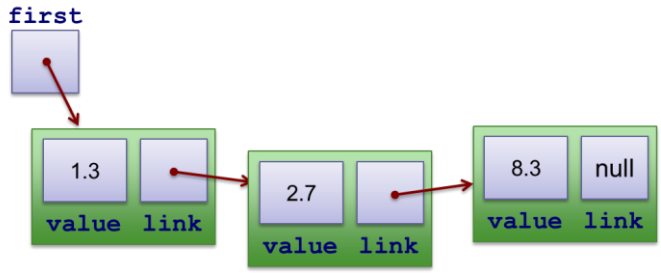
```
void strCat(char frontStr[], char endStr[]) {
```

c) Implement the function `newRepeatedStr`, which returns a new dynamically allocated string which is the same as `origStr` repeated `repeat` number of times. For example, if `origStr` is "abcd" and `repeat` is 3, the new string "abcdabcdabcd" is returned. (*Hint*: you will need to use a separate loop to first determine the size of the string `origStr`.)

```
char* newRepeatedStr(char *origStr, int repeat) {
```

6. a) Using the given struct definition for doubleNode, create the linked-list shown below:

```
typedef struct _doubleNode {
    double value;
    struct _doubleNode* link;
} doubleNode;
```



b) Now, write the code that will free up the memory allocated for this linked-list.

7. What is the main difference between how memory that is allocated using malloc, and memory that is allocated when a (non-pointer) variable is declared, handled in C?

Part 2 – Old Stuff

8. Write functions that take a non-empty array of doubles, and its length as arguments, and returns:

a) The sum of the items

b) The index of the maximum value

c) A boolean which indicates if the numbers are in strictly increasing order

9. Implement a recursive function that returns the number of times the digit 3 will appear in the octal representation of a given positive integer n.

10. Implement the following function which computes the given summation using only a single loop. You may assume input values for n will always be positive.

$$1x^1 + 2x^2 + 3x^3 + \cdots + nx^n$$

```
double polySum(int n, double x) {
```